# Choosing a Ratio Representation
## Answer Key

Grade 6–7    Ratios and Equivalent Ratio Tables

**Quick Answers**

| | | | |
|---|---|---|---|
| 1. 4, 6, 8 | 4. 12 | 7. 10 | 10. 5 |
| 2. 2.5 | 5. — | 8. 35 | 11. — |
| 3. 28, 42, 70 | 6. 29 | 9. — | 12. —, 14, 9 |

**Curriculum**

**Level:** Grade 6–7    Ratios and Equivalent Ratio Tables
6.RP.A.2, 6.RP.A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
*Difficulty 1–5 of 5 across 18 tagged checks.*

**Common wrong answers**

7. *If they got 7: added the same amount to both parts of the ratio instead of multiplying both by the same factor*

### 1. Trail mix, 3 cups oats to 2 cups raisins — complete the table, then find the raisins for 12 cups of oats.

**Step 1.** A table is the right tool because the question asks for several equivalent ratios at once, not just one. Each new column comes from multiplying *both* parts by the same number.
**Step 2.** 6 cups of oats is $3 \times 2$, so the raisins double: $\boxed{4}$
**Step 3.** 9 cups of oats is $3 \times 3$, so the raisins triple: $\boxed{6}$
**Step 4.** 12 cups of oats is $3 \times 4$, so multiply the raisins by four: $\boxed{8}$
*What to look for:* the student should name the multiplier ($\times 4$ for the last column), not just fill in the number.

### 2. Five notebooks for 12.50 dollars — cost of one.

**Step 1.** The question asks for the cost of a single item, which is the definition of a unit rate: divide the total by the number of items.
**Step 2.** $12.50 \div 5 = 2.50$ dollars each. $\boxed{2.50}$
*Why a unit rate here:* once you know the price of one, any quantity is a single multiplication — which is exactly what problem 5 will need.

### 3. Printer: 14 pages every 4 minutes — complete the table.

**Step 1.** Every column is the same ratio, so find the multiplier from the minutes row and apply it to the pages row.
**Step 2.** 8 minutes is $4 \times 2$, so the pages double: $\boxed{28}$
**Step 3.** 12 minutes is $4 \times 3$: $\boxed{42}$
**Step 4.** 20 minutes is $4 \times 5$: $\boxed{70}$
*Alternative route:* the unit rate is $14 \div 4 = 3.5$ pages per minute, and $3.5 \times 20 = 70$. Both representations are correct — the table is faster for the easy multiples, the unit rate for the awkward ones.

### 4. Map: 4 cm represents 5 km. How many cm for 15 km?

**Step 1.** One unknown value and one known ratio is exactly the shape a proportion handles, so set the two ratios equal.
**Step 2.** $\frac{x}{15} = \frac{4}{5}$. Multiplying both sides by 15 gives $x = 15 \cdot \frac{4}{5}$.
**Step 3.** $15 \div 5 = 3$, and $3 \times 4 = 12$. $\boxed{x = 12}$
*Check:* 12 cm to 15 km reduces to 4 cm to 5 km, the given ratio.

### 5. Better buy: 5 for 12.50 dollars, or 6 for 17.40 dollars?

**Step 1.** The two offers have different pack sizes, so they cannot be compared as they stand. Put both in the same form — cost per notebook.
**Step 2.** Corner store: $12.50 \div 5 = 2.50$ dollars each. Supermarket: $17.40 \div 6 = 2.90$ dollars each.
**Step 3.** Written as a comparison of the two rates: $\boxed{2.50 < 2.90}$
The corner store is the better buy: 2.50 per notebook is less than 2.90.
*Common wrong answer:* the supermarket, from noticing it sells more notebooks. More items for more money says nothing until both are per-one.

### 6. 174 miles on 6 gallons — miles per gallon.

**Step 1.** "Per gallon" names the unit: divide the miles by the gallons.
**Step 2.** $174 \div 6 = 29$. $\boxed{29}$
*Check by scaling back:* $29 \times 6 = 174$, the distance given.

### 7. Sam extended 2 to 5 by writing 4 to 7. Fix it.

**Step 1 (a).** Equivalent ratios are made by *multiplying* both parts by the same number, not by adding. Adding 2 to both parts changes how many times bigger one part is than the other: 5 is 2.5 times 2, but 7 is only 1.75 times 4. The mix would come out weaker.
**Step 2 (b).** From 2 cups of concentrate to 4 cups is a multiplier of 2, so the water must also double: $5 \times 2$. $\boxed{10}$
*Common wrong answer:* 7 — the additive mistake this problem is built to catch. A quick test: does the ratio 4 to 7 reduce to 2 to 5? It does not.

### 8. Soup: 8 servings uses 20 oz. How many oz for 14 servings?

**Step 1.** 14 is not a whole-number multiple of 8, so a table would take several awkward steps — a proportion handles it in one.
**Step 2.** $\frac{x}{14} = \frac{20}{8}$, so $x = 14 \cdot \frac{20}{8} = 14 \cdot 2.5$.
**Step 3.** $14 \times 2.5 = 35$ ounces. $\boxed{x = 35}$
*Same answer, other route:* the unit rate is $20 \div 8 = 2.5$ ounces per serving.

### 9. Runner A: 9 km in 40 min. Runner B: 7 km in 32 min. Who is faster?

**Step 1.** Neither the distances nor the times match, so a direct comparison is impossible until both are in the same form. Kilometres per minute works.
**Step 2.** A: $9 \div 40 = 0.225$ km per minute. B: $7 \div 32 = 0.21875$ km per minute.
**Step 3.** Comparing the two rates: $\boxed{0.225 > 0.21875}$
Runner A is faster — more kilometres in each minute.
*Alternative representation:* minutes per kilometre works too, but then the *smaller* number wins (4.44 against 4.57). Say which form you used before deciding which direction is better.

### 10. 3 gallons cover 1050 sq ft. How many gallons for 1750 sq ft?

**Step 1.** 1750 is not a neat multiple of 1050, so go through the unit rate: coverage per one gallon.
**Step 2.** $1050 \div 3 = 350$ square feet per gallon.
**Step 3.** $1750 \div 350 = 5$ gallons. $\boxed{x = 5}$
*Check:* $5 \times 350 = 1750$, the coverage needed.

### 11. Team A won 6 of 9; Team B won 10 of 15. Better ratio?

**Step 1.** The two teams have played different numbers of games, so the win counts cannot be compared directly. Reduce each ratio to lowest terms.
**Step 2.** 6 to 9 divides by 3, giving 2 to 3. 10 to 15 divides by 5, also giving 2 to 3.
**Step 3.** The two ratios are the same, so neither team is ahead: $\boxed{6/9 = 10/15}$
*What makes this hard:* Team B has won more games, so the intuitive answer is Team B. Reducing both ratios is what settles it — both teams win two of every three.

### 12. Challenge: Recipe R is 2 concentrate to 5 water; Recipe S is 3 concentrate to 8 water.

**Step 1 (a).** "Stronger" means more concentrate for the same amount of water, so compare concentrate per cup of water: $2 \div 5 = 0.4$ and $3 \div 8 = 0.375$.
**Step 2 (a).** Recipe R carries more concentrate in every cup of water, so it is the stronger mix. Written as a comparison of the two ratios: $\boxed{2/5 > 3/8}$
**Step 3 (b).** Recipe R: 35 cups of water is $5 \times 7$, so multiply the concentrate by seven as well: $2 \times 7$. $\boxed{14}$
**Step 4 (c).** Recipe S: 24 cups of water is $8 \times 3$, so $3 \times 3$. $\boxed{9}$
*Why three different tools appear in one problem:* part (a) needs a comparison of two rates, parts (b) and (c) need a table-style scale-up, and each is chosen because of the numbers in front of you — which is the whole point of the sheet.